DE–7230 **11**

DISTANCE EDUCATION

M.Sc. (C.S.) DEGREE EXAMINATION, DECEMBER 2013.

MATHEMATICAL FOUNDATION OF COMPUTER SCIENCE

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($10 \times 3 = 30$ marks)

Answer ALL the questions.

1. What is conjunction? Give an example.

2. What is tautology? Give an example.

3. Define Complementary relation.

4. What is chain?

5. What is surjection? Give an example.

6. Define Inverse function.

7. Define Monoid.

8. What is abelian group? Give an example.

9. What is cycle? Give an example.

10. What is adjacent matrix?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Construct the truth table for
$(P \cap Q) \vee (\neg P \cap Q) \vee (P \cap \neg Q) \vee (\neg P \cap \neg Q)$.

12. Show that
$(x)(P(x) \to Q(x)) \cap (x)(Q(x) \to R(x)) \Rightarrow (x)(P(x) \to R(x))$

13. Let R be the relation represented by the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find the matrix representing

(a) R^2

(b) R^3.

14. Show that the function $f(x) = x3$ and $g(x) = x\,1/3$ for $x \in R$ are inverse of one another.

15. Show that $(S,+)$ and $(P,*)$ are isomorphic.

16. A graph G with P points and $\delta \geq \frac{P-1}{2}$ is connected.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Obtain the principal disjunctive adconjunctive normal forms $(P \to (Q \cap R)) \cap (\neg P \to (\neg Q \cap \neg R))$.

18. Prove that $\psi_{A \cup B}(x) = \psi_A(x) + \psi_B(x) - \psi_{A \cap B}$ where +, – are usual arithmetic operations.

19. Let $(H,*)$ be a sub group of $(G,*)$. The set of left cosets of H in G form a partition of G. To prove every element of G belongs to one and only one left cost of H in G.

DE–7231 **12**

DISTANCE EDUCATION

M.Sc. (CS) DEGREE EXAMINATION, DECEMBER 2013.

COMPUTER ARCHITECTURE

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are different sizes of operands?

2. State the principle of encoding an instruction set.

3. What is meant by instruction level parallelism?

4. What is linear pipeline processor?

5. Define : Fine grain parallelism.

6. What are vector pipelines?

7. Define the terms, "Hit ratio" and "Miss rate" related to cache memory.

8. List down the types of storage devices.

9. Define the term, "multithreading" with respect to parallel system.

10. What are performance issues occur in a distributed shared memory systems?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. List and explain any five types of addressing modes with illustration.

12. Write the concepts of ILP in detail.

13. Give a brief account on software speculation for a parallel system.

14. Discuss the concept of RAID concept related to memory.

15. Narrate the importance of main memory performance and its technology.

16. Describe the synchronization models with its memory consistency.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. With neat diagram, explain the TM32 architecture in detail.

18. Give a detailed description on advanced compiler support for VLIW architecture system.

19. Describe the concepts of multiprocessors and thread level parallelism.

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DE–7232 **13**

DISTANCE EDUCATION

M.Sc.(CS) DEGREE EXAMINATION, DECEMBER 2013.

DATA STRUCTURES USING C++

(2007 Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions

1. Define an array.
2. How the classes are defined in c++?
3. What is meant by a stack?
4. Mention the limitations of the program.
5. What is meant by Linear Data Structure?
6. Give an example for circular Linked List.
7. Define Tree.
8. How to destroy a binary search tree?
9. What is meant by SORTING?
10. How Merge sort differs from quick sort?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. How to use two dimensional arrays?
12. Explain the concept of overloading functions.

13. How queue can be represented sequentially?

14. Write a program in C++ to evaluate a post fix expression.

15. Mention any one application of Linked List and discuss it.

16. How to create a binary search Tree?

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Discuss on the different types of Inheritance supported in C++.

18. Explain the following tree traversals:

    (a) Inorder

    (b) Preorder

    (c) Post order.

19. Describe the technique for Bubble sort.

DE–7233 **14**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2013.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. List any three header files in Windows programming.
2. Define Device Context.
3. Write any three data types in VB.
4. What is meant by Modules in VB?
5. Write down the difference between Check box and Option button controls in VB.
6. What is connection object?
7. What are Accelerators?
8. Differentiate between Modal and Modaless dialog.
9. What do you mean by OLE?
10. Write a note on Open Data Base Connectivity.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain in detail about Windows Message structure and Windows procedures.
12. Explain the Looping constructs in VB.
13. List the properties of Recordset object.

14. Detail on Menus and Dialog boxes in VC++.

15. Explain the Document view architecture in detail.

16. Detail on Exception handling and Debugging steps in VC++.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. With the help of a flow chart. Explain how a Windows program works.

18. Compare and Contrast the following :

    (a) Combo box and Data Combo control.

    (b) List box and Data List control.

    (c) DAO and ADO.

19. Explain the following :

    (a) Serialization.

    (b) Files in VC++

    (c) OLE.

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DE–7234 **15**

DISTANCE EDUCATION

M.Sc. (CS) DEGREE EXAMINATION, DECEMBER 2013.

DATABASE MANAGEMENT SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)
Answer ALL questions.

1. Distinguish schema from subschema.

2. What is meant by conceptual view of database?

3. Give the syntax for select operation. Also illustrate.

4. Write about natural join operation.

5. What is meant by rollback segment?

6. Define :
   (a) System monitor
   (b) Process monitor

7. Write the general form SQL operation, “rename”.

8. Mention five built in SQL aggregate functions.

9. How will you create a table in oracle?

10. What is a dynamic SQL?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Narrate the advantages of Data Base scheme.

12. Illustrate and explain the Cartesian product operation.

13. Discuss the concept of oracle concurrency control scheme.

14. Give a note on oracle client/server system.

15. Discuss the SQL set operations with suitable examples.

16. Summarise the role of oracle precompiler in embedded SQL.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. With neat diagram, explain the oracle database architecture.

18. List down various nested subqueries followed in SQL and explain with suitable examples.

19. Explain the procedure of creating, modifying, renaming, copyright and dropping a table with suitable illustration.

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DE–7235 **21**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2013.

COMPUTER NETWORKS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Explain any one uses of computer networks.
2. Write notes on transmission media.
3. What is meant by petrinet model?
4. Give details about error detecting codes.
5. Mention the role of firewalls.
6. What is an IP Address?
7. Write note on quality of service parameters of transport layer.
8. What is UDP?
9. What are substitution ciphers?
10. What is meant by JPEG standards?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Compare virtual circuits versus circuit switching.
12. Write a detail note on a simplex stop-and-wait protocol.
13. Explain various services provided to transport layer.

14. Discuss about upward and downward multiplexing.

15. Give a detail note on digital signatures.

16. Discuss about Internet transport protocols.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write a detail note on OSI reference model?

18. What is meant by Routing? Explain shortest path routing algorithm.

19. Give a brief note on electronic mail.

DE–7236 **22**

DISTANCE EDUCATION

M.Sc. (CS) DEGREE EXAMINATION, DECEMBER 2013.

SOFTWARE ENGINEERING

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Write a note on: Software Engineering Layers.
2. Distinguish between Direct and Indirect measures of software Engineering Process.
3. What are project resources?
4. Write a note on: Risk Identification.
5. Write about Task sets.
6. Write the steps for statistical software quality assurance.
7. What is requirement analysis?
8. Why is Architecture Important?
9. Define: White Box Testing.
10. Write a note on: Top-down Integration Testing.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write short notes on the Liner sequential Model.
12. Discuss briefly on: Fourth Generation Techniques.
13. Explain about Risk Projection.

14. Write short notes on: Formal Technical Reviews.

15. Explain the system engineering hierarchy.

16. What is basis path testing? Explain.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain about Evolutionary software process models.

18. Describe in detail, Empirical Estimation Models.

19. Discuss the following:

(a) SQA Activities. (8)

(b) System testing. (7)

DE–7237 **23**

DISTANCE EDUCATION

M.Sc. (Comp. Science) DEGREE EXAMINATION,
DECEMBER 2013.

INTERNET PROGRAMMING AND WEB DESIGN

(2007 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is decryption?

2. Write any three internet programming languages.

3. Write note on CGI.

4. What is a scripting languages?

5. What are the attributes associated with font lag?

6. Write any two benefits of using SGML.

7. Write the functions of VDO live system.

8. What is web POP?

9. Write the fundamental kinds of websites.

10. Write any two animation techniques.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain firewall with its types.

12. How java script works in HTML documents. Explain with an example.

13. Write a procedure to edit map files on the mac.

14. Explain Netsape extensions.

15. Explain active X control with example.

16. How do you create and implement a websites.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. How can you declare and import subroutines in perl? Explain with example.

18. Describe the procedure for adding objects to web pages and how can you link VB soupl with objects.

19. Write a HTML program using any scripting language.

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DE–7238 **24**

DISTANCE EDUCATION

M.Sc.(CS) DEGREE EXAMINATION, DECEMBER 2013.

IMAGE PROCESSING AND ANALYSIS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is meant by sampling and quantization?

2. Define the term industrial automation.

3. What are the types of image enhancement techniques?

4. What is the use of smoothing operation?

5. Define signal-to-noise ratio.

6. What are the two major steps in Huffman encoding scheme?

7. Define the term edge detection.

8. Define the term image segmentation.

9. What is critical pixel and end pixel?

10. Define the term area of an object.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write notes on visual detail in the digital image.

12. Discuss about linear and non-linear stretching with example.

13. Explain vector quantization compression.

14. Explain the stereo imaging algorithm to recover depth.

15. Discuss about corner detection.

16. Explain tree search with A* algorithm.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain in detail the components of image processing system.

18. Describe in detail the processing of colour images.

19. Discuss about geometrical attributes in feature extraction.

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DE–7239 **25**

DISTANCE EDUCATION

M.Sc.(CS) DEGREE EXAMINATION, DECEMBER 2013.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. What are Real Time Systems?
2. What are operating system components?
3. What are Threads?
4. Define: Semaphores.
5. Write a note on: Swapping.
6. Define: Virtual Memory.
7. What is a file system?
8. Write a note on: Disk structure.
9. What are Distributed system?
10. List the features of LINUX OS.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain any TWO CPU Scheduling Algorithms.
12. Write about the critical Section Problem.

13. Discuss briefly Recovery from Deadlock.

14. Write short notes on: Paging.

15. Discuss about File concepts and Access Methods.

16. Explain Statefull versus stateless services.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write short notes on: (8+7)

(a) Desktop systems

(b) Multiprocessor systems.

18. Explain about Inter Process Communication.

19. Discuss the following: (7+8)

(a) Demand Paging.

(b) Design Issues in Distributed Operating System.

DE–7648 **16**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2013.

C++ AND DATA STRUCTURES LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks

Record : 10
Algorithm : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

Examiner has to select and give ONE question to each candidate by LOT system.

1. (a) Write a C++ program to convert a give number into words from 1 to 5.

   (b) Write a C++ program to sort a set of elements using Selection Sort.

---------- Cut here ----------

2. (a) Write a C++ program to find the roots of the quadratic equation $ax^2 + bx + c = 0$.

   (b) Write a C++ program to sort a set of elements using Insertion sort.

3. (a) Write a C++ program to find the roots of a quadratic equation using function.

   (b) Write a C++ program to sort a set of elements using Quick Sort.

---------- Cut here ----------

4. (a) Write a C++ program for multiplication of two matrices.

   (b) Write a C++ program for Linked List implementation of Queue operations.

---------- Cut here ----------

5. (a) Write a C++ program to find the factorial of a given number using recursion.

   (b) Write a C++ program to implement push and pop operations on stack.

---------- Cut here ----------

6. (a) Write a C++ program to concatenate two given strings (using dynamic constructor).

   (b) Write a C++ program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

7. (a) Write a menu driven C++ program to add and subtract given two matrices of order $m \times n$ defined in class, using operator overloading.

   (b) Write a C++ program to implement insert and delete operations on queue using array concept.

8. (a) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

   (b) Write a C++ program to implement push and pop operations on stack.

---------- Cut here ----------

9. (a) Write a C++ program to sort ten numbers in ascending order with naming of variable and the value before and after sorting.

   (b) Write a C++ program to read TEN values to an array variable. Use pointers to locate and display each value.

DE–7649 **17**

DISTANCE EDUCATION

M.Sc. (CS) DEGREE EXAMINATION, DECEMBER 2013.

VISUAL PROGRAMMING LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by lot system.

Break up of mark :

Record : 10
Algorithm: 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

1. (a) Using Active X control create a text box that accepts only numeric value with the following properties.

   (i) Background of the text box

   (ii) Foreground of the text box

   (iii) Text property of the text box

   (iv) Resize the text box at the standard application without using properties

   (b) Write visual C++ program to fill background of the client area with a bitmap.

2. (a) Using AxDLLor EXE add a class module that would perform the following function
   (i) Text whether the given number is perfect or not
   (ii) Whether the given number is Armstrong number or not
   (iii) Find the factorial of the given number
   (iv) Sum of digits.

   (b) Write a visual C++ program to generate a status bar and show the status of caps lock, Num lock and Scroll lock in it

---------- Cut here ----------

3. (a) Write a VB program to perform the following operation in a record of random Access file
   (i) Insert
   (ii) Delete
   (iii) Search.

   (b) Write a visual program to create a window of desired size using MFC.

---------- Cut here ----------

4. (a) Create an employee with Empno, Empname, Basic pay, HRA, DA, PF, LIC, GP and NP with the following calculation.

   HRA = 10% OF BP

   PF = 3% OF BP

   GP = BP + DA + HRA

   DA = 5% OF BP

   LIC = 5% OF BP

   NP = GP –(PF + LIC)

   Using REMOTE DATA ACCESS OBJECT, implement the following operations :
   (i) Insert a record
   (ii) Search and delete a record
   (iii) Modify the record
   (iv) Display all the employee records whose names are starting with the letter "S"

   (b) Write a visual C++ program to create a list box in a window.

5. (a) Using built in Ax control, develop the windows NOTEPAD with File and Edit menu operation and also display the floating menu. Whenever necessary (use Rich Text Box, menu editor)

(b) Write a VC++ program to find out whether a mouse is attached or not; and if attached. How many buttons are present or not.

---------- Cut here ----------

6. (a) Create a table hospital with the following fields Patient no, Patient Name, address, blood group, Diseases using Ax data access object, develop a hospital management system with the following operation.

(i) Insert a record into the table

(ii) Search and delete a record

(iii) Modify the record

(iv) Display all the patients details with the corresponding blood group from the Combo Box.

(b) Write a visual C++ program to create a window of desired size using MFC.

---------- Cut here ----------

7. (a) Create a VB application with a Drive List box, Dir List box

(i) Select exe file and execute it

(ii) Select the picture and load it to the form

(iii) Filter the file in the file list box according to the extension chosen in the combo box.

(b) Write a visual C++ program to handle windows messages in MFC program.

8. (a) Using Msflex grid control, display the multiplication and addition table of 20 rows and columns (use MSFlex, Grid, Button)

(b) Write a visual C++ program to get the status of the shift and toggle keys. Using MFC.

---------- Cut here ----------

9. (a) Develop a Data report using Employee Table with the following.

(i) Display all the employee details

(ii) Display all the employee details in each dept and display total salary in each dept.

(iii) Display all the employee details which starts with the employee name specified in Text box.

(b) Write a visual C++ program to create a list box in a window.

---------- Cut here ----------

10. (a) Using a control array, create a simple calculator which will do the following operations

(i) Addition

(ii) Subtraction

(iii) Multiplication

(iv) Division

(v) Square

(vi) Square root

(vii) Modulus

(viii) Power

(ix) Percentage

(Using text box, command button, frames)

(b) Write visual C++ program to fill background of the client area with a bit map.

DE–7650 **26**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2013.

LAB III – INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Break up marks

Record	: 10
Algorithm	: 10
Program	: 50
Debugging	: 10
Execution	: 10
Result	: 10
Total	: 100

Examiner has to select and give one question to each candidate by lot system.

1. (a) Using Javascript create a Multiplication table.

   (b) Design a registration form using HTML for participating in a symposium.

---------- Cut here ----------

2. (a) Write a Java Program to illustrate Multithreading.

   (b) Write a swing program to create the Tabbed Panels.

---------- Cut here ----------

3. (a) Design a webpage for a traveling agency.

   (b) Write a Java Program to create a color as the background.

4. (a) Develop a picture gallary having atleast 5 pages. Each of them is having several pictures with suitable information.

   (b) Write a Java program using applet to display the different colors and fonts.

---------- Cut here ----------

5. (a) Write a Java Program using Applet to create the frames and its controls.

   (b) Design and publish a webpage for a college.

---------- Cut here ----------

6. (a) Write a menu driven Java Program to do database functions using database connectivity facility. (Edit, Insert and Delete)

   (b) Write a swing program to create buttons with

      (i) Tool tip text

      (ii) Image

      (iii) Border

      (iv) Shortcut Key

---------- Cut here ----------

7. (a) Write a VB script to do the following.

      (i) Check the given password

      (ii) Change the existing password.

   (b) Write a Java Program using Applets to display the different colours and fonts.

8. (a) Write a VB Script to create a calender for given month and year.

(b) Develop a picture gallery having atleast 5 pages. Each of them is having several pictures with suitable information.

---------- Cut here ----------

9. (a) Write a Java Program using Applet to display dialogue menu in applet.

(b) Design and publish webpage for a college.